

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1253

High Falls - Croghan Ny 100 kV
January 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1253. The Interconnection Customer proposes a 285.4 MW solar plus storage facility interconnecting at the High Falls - Croghan Ny 100 kV substation in the Indiana area. The request was received on July 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 398 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,368,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Twin Branch Pv to Scenic Woods 100 kV line is overloaded to 139.1% of its 170 MVA emergency rating for the loss of the Kemper County to Twin Branch 100 kV. The project contributes 36.2% DFAX (103.3 MW) to this overload.

Scenic Woods to High Falls - Croghan Ny 100 kV: 135.8% loading (143 MVA) following the outage of the Twin Branch to Kemper County 100 kV; DFAX = 36.2%.

The Parkrose to Twin Branchpv 100 kV is overloaded to 112.6% of its 191 MVA emergency rating for the loss of the Twin Branch to Kemper County 100 kV L. The project contributes 36.2% DFAX (103.3 MW) to this overload.

N Mesquite to Denver Terminal 100 kV line: 110.9% loading (144 MVA) following the outage of the Scenic Woods to Twin Branch Pv 100 kV; DFAX = 16.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Twin Branch Pv to Scenic Woods 100 kV line	Rebuild line with higher-capacity conductor	\$1,446,000	35% (\$504,000)
Scenic Woods to High Falls - Croghan Ny 100 kV	Rebuild line with higher-capacity conductor	\$1,079,000	34% (\$372,000)
Parkrose to Twin Branch Pv 100 kV line	Rebuild line with higher-capacity conductor	\$5,453,000	35% (\$1,901,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Denver Terminal to N Mesquite 100 kV	Rebuild line with higher-capacity conductor	\$2,407,000	24% (\$591,000)
	Total allocated to project		\$3,368,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.